Parul® University
Vadodara, Gujarat

NAAC GRADE A++

Department of Computer Science and Engineering

Deep Learning with NLP

Study Guide

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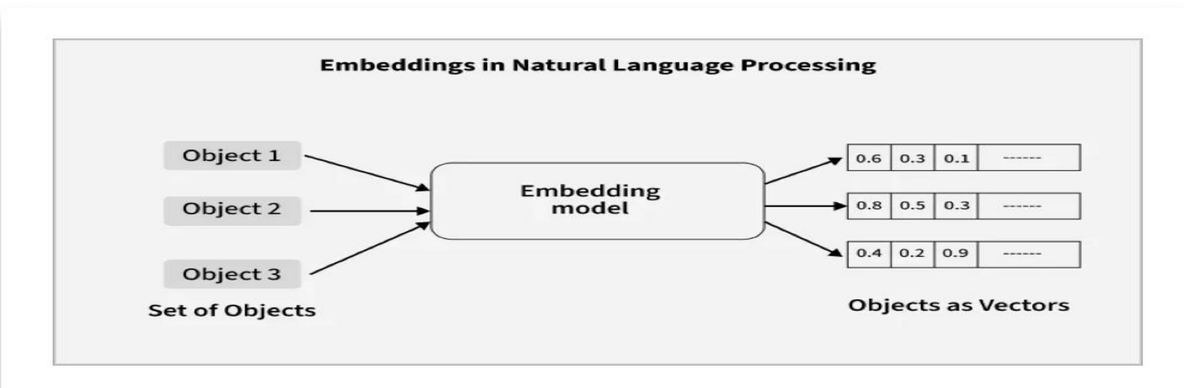
UNIT 6: Word Vector Representation

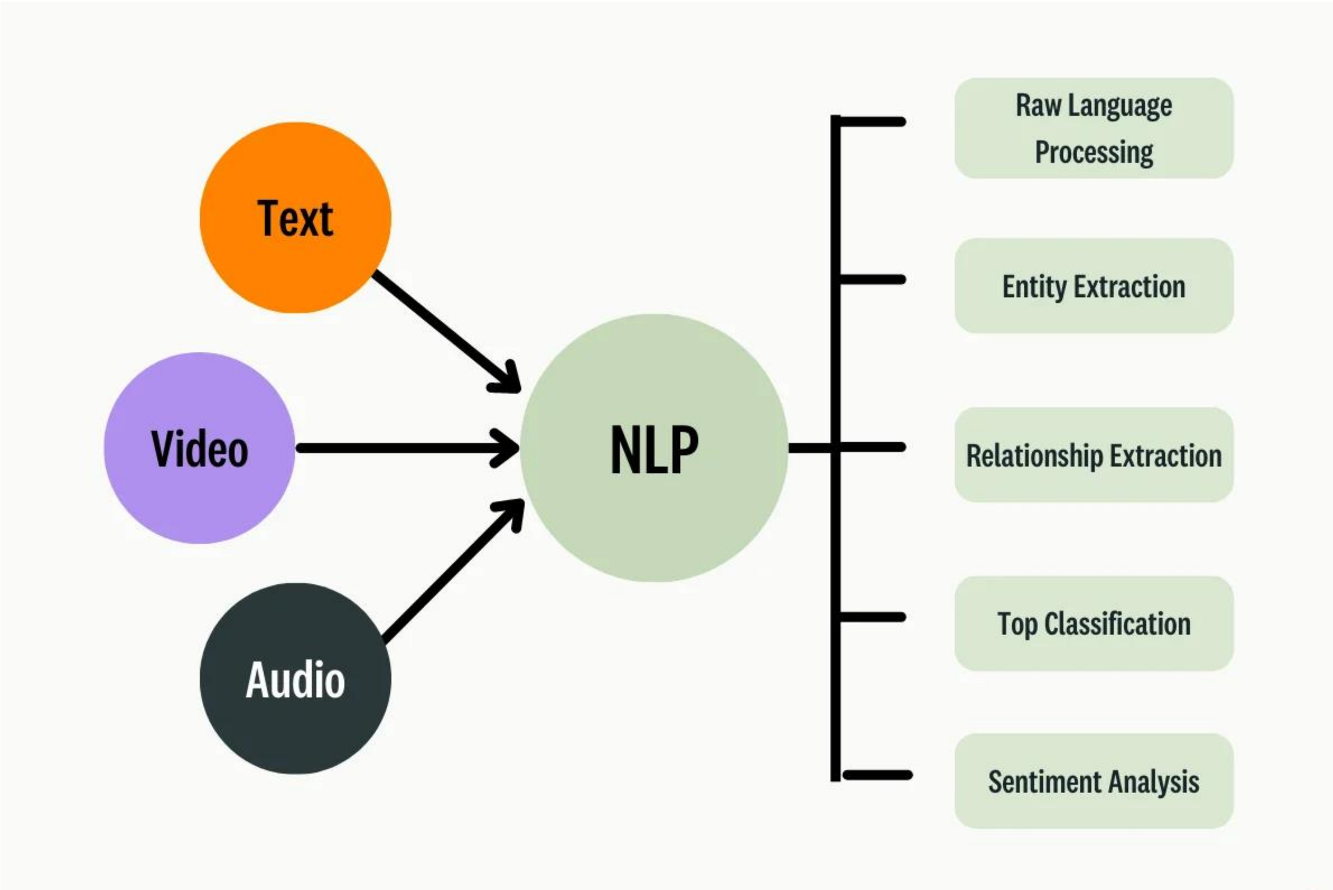
INTRODUCTION TO VECTOR REPRESENTION

Word Embeddings are numeric representations of words in a lower-dimensional space, that capture semantic and syntactic information. They play a important role in Natural Language Processing (NLP) tasks. Here, we'll discuss some traditional and neural approaches used to implement Word Embeddings, such as TF-IDF, Word2Vec, and GloVe.

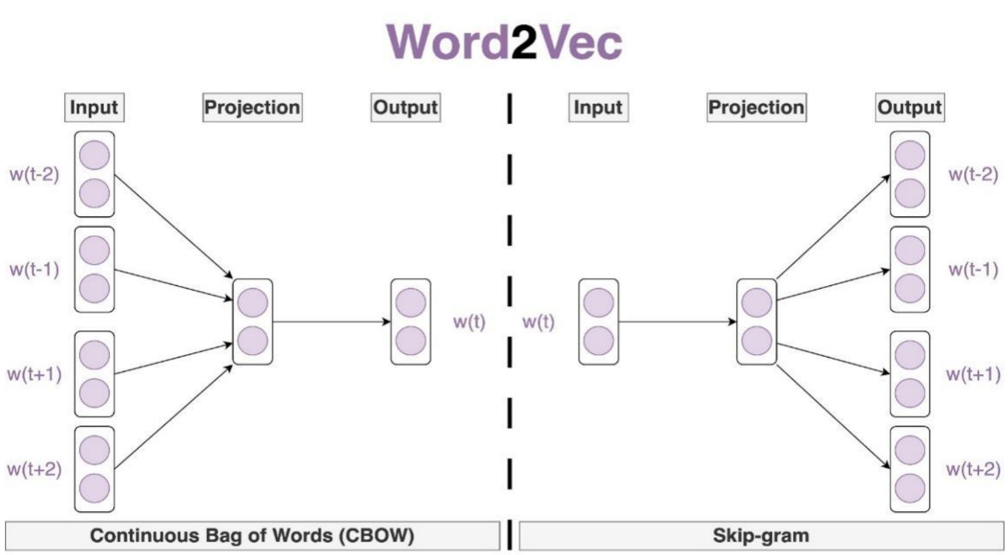


Neural Language Model

Natural Language Processing (NLP) helps machines to understand and process human languages either in text or audio form. It is used across a variety of applications from speech recognition to language translation and text summarization.


Text
Video
Audio
NLP
Raw Language Processing
Entity Extraction
Relationship Extraction
Top Classification
Sentiment Analysis

Word2Vec


Word2Vec
Input
Projection
Output
Input
Projection
Output
w(t-2)
w(t-1)
w(t+1)
w(t+2)
w(t)
w(t)
w(t-2)
w(t-1)
w(t+1)
w(t+2)
Continuous Bag of Words (CBOW)
Skip-gram

Skip-Gram Model

The skip-gram model was introduced by Mikolov et al. in their paper "Efficient Estimation of Word Representations in Vector Space" (2013). The skip-gram model is a way of teaching a computer to understand the meaning of words based on the context they are used in. An example would be training a computer to understand the word "dog" by looking at sentences where "dog" appears and seeing the words that come before and after it. By doing this, the computer will be able to understand how the word "dog" is commonly used and will be able to use that understanding in other ways.

Hidden Layer

Hidden layers are the intermediate computational tiers in an artificial neural network situated between the input layer and the output layer. They act as the "brain" of deep learning, performing complex mathematical transformations that allow the AI to recognize intricate patterns and abstract features in data.

CBOW Model

Continuous Bag of Words (CBOW) is a popular natural language processing technique used to generate word embeddings. Word embeddings are important for many NLP tasks because they capture semantic and syntactic relationships between words in a language.

what is Negative Sampling ?

When working with natural language processing (NLP) tasks, especially word embeddings, efficiently optimizing models like Word2Vec becomes critical due to the vast vocabulary sizes involved. Negative Sampling is a powerful technique that simplifies and accelerates the training of such models by approximating the computationally expensive softmax function.

Sampling-based approximations for Softmax

Main Sampling-Based Softmax Approximations
1. Negative Sampling
Introduced in:
Efficient Estimation of Word Representations in Vector Space
Used heavily in:
word2vec

Glove, Cross-lingual word embedding models
Properties of GloVe Embeddings
Semantic Regularities
Famous analogy behavior:
king−man+woman≈queen
king−man+woman≈queen
because linear relationships emerge naturally.